

# When usage exceeds capacity

Each GCell edge has capacity two on tiny\_gr

# The idea

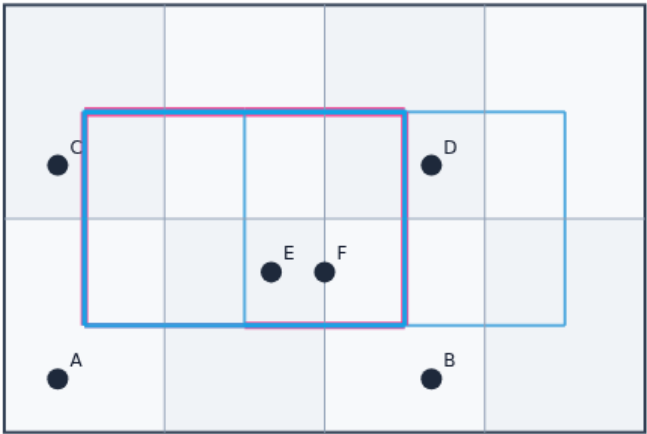
- For each edge  $e$ , overflow  $e$  equals  $\max(0, \text{usage } e - \text{capacity } e)$
- Total is the sum across edges
- Max is the worst edge
- Count is how many edges have positive overflow
- Sequential L-HV on all six nets should yield positive total overflow at cap two

# Overflow

EDGE OVERFLOW

## Overflow

Step 1 / 5 · def · module03-01-edge-overflow



### Explanation

$ov(e) = \max(0, usage(e) - Cap)$  per edge.

- Not tile RUDY
- Router metric

Edge-local

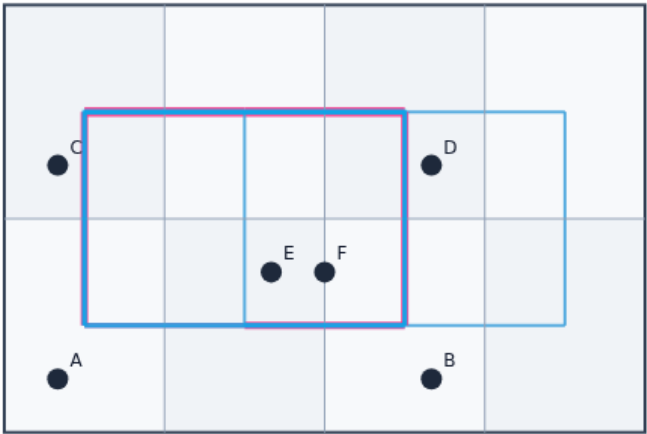
Overflow

# Total overflow

EDGE OVERFLOW

## Total overflow

Step 2 / 5 · total · module03-01-edge-overflow



### Explanation

Sum over edges—primary regression scalar.

- Sum  $ov(e)$
- Report every pass

```
spread L-HV≈2
```

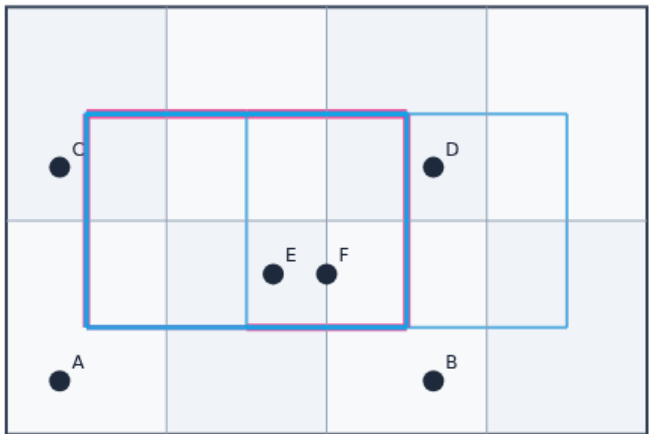
Total overflow

# Max overflow

EDGE OVERFLOW

## Max overflow

Step 3 / 5 · max · module03-01-edge-overflow



### Explanation

Worst edge—catches hotspot corridors.

- Hot corridor
- max metric

max≈1

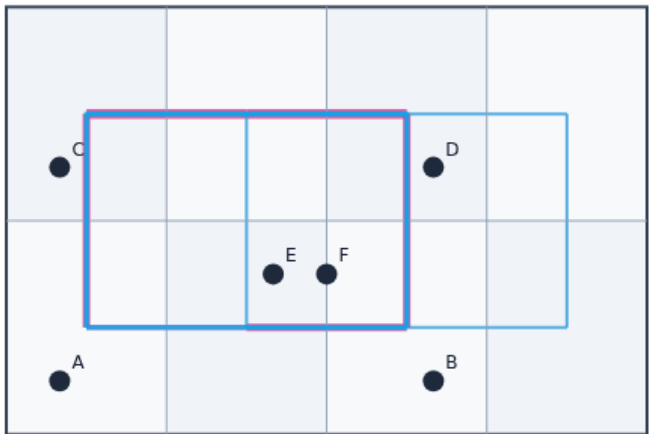
Max overflow

# Overflow count

EDGE OVERFLOW

## Overflow count

Step 4 / 5 · count · module03-01-edge-overflow



### Explanation

How many edges exceed Cap.

- Triple report
- Same idea as congestion labs

count edges

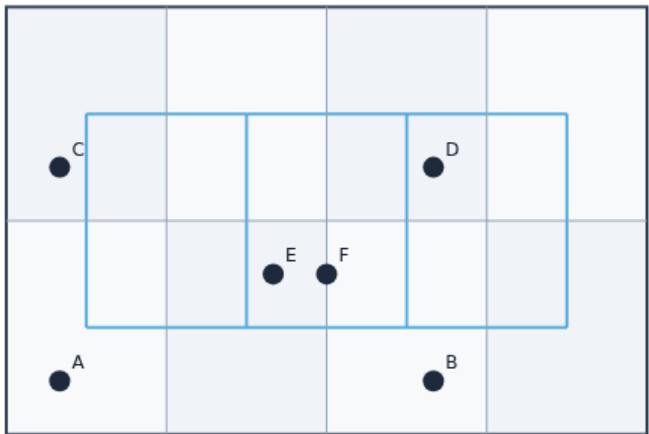
Overflow count

# Hit targets

EDGE OVERFLOW

## Hit targets

Step 5 / 5 · target · module03-01-edge-overflow



### Explanation

Move pins or switch HV/VH/maze to clear thresholds.

- Check scores learner
- No golden mode

Browser challenges

Hit targets

# Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Edge overflow metrics

INTERACTIVE TOOL

## Edge overflow metrics

Report total, max, and overflowing-edge count from routed usage.

**Your job:** route L-HV on spread (total  $\approx 2$ ), then hit overflow targets by moving cells or switching L-VH / maze.

Reset workspace

Challenges0 / 8 cleared

Pick one[Intro] Spread L-HV overflow

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)  
routes: (none — click Route)  
overflow total: 0.00 · max: 0.00 · count: 0  
edge capacity: 2  
HPWL: 34.0  
view: your placement

Browser lab starter

learn\_global\_routing — 01 edge overflow



# Implement track

- Implement ``edge_overflow(usage, capacity)``
- Call `route_nets` with mode `I_hv` on `tiny_gr` and assert total overflow is greater than zero

# Pitfalls

- Computing overflow before summing all nets
- Using tile demand from congestion instead of edge usage
- Reporting negative overflow values

# Your turn

- Hit overflow targets
- Next: rip-up the hottest net and maze reroute